Shiye Du

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Education

Westlake University Hangzhou, China
Bachelor of Science in Chemistry **July 2023 - Present**

- **Rank:** 1/89
- **GPA:** 4.070/4.3
- **Coursework:** Organic Chemistry, Inorganic Chemistry, Physical Chemistry, Fundamental Molecular Spectroscopy and Group Theory, Bioconjugation Technique, Inorganic Chemistry Characterization Methods.

Duke University Durham, United States
Aug. 2025 - Dec. 2025

- **Coursework:** Biochemistry, Data Science and Statistics, Discrete Math for CS, Elementary Differential Equation.

Research Experience

Professor Hongyu Chen's Lab, Westlake University Hangzhou, China
Undergraduate Researcher **Jan. 2025 - Present**

Project: Universal Nanoparticle Descriptors for Physics-Aware Inverse Design

- Built a descriptor set for colloidal nanoparticle assemblies, including hot-spot volume and intensity, normal electric-field volume, global/component complex multipolar amplitudes, and derived post-processed descriptors.
- Established these physics-informed descriptors as intermediate design variables to improve generalizability, interpretability, and stability beyond direct spectrum-to-structure inverse mapping in complex nanostructures.

Project: Optical Force and Torque Enhancement in Coupled Plasmonic Systems

- Performed high-throughput screening of ring, shell, and rod configurations to identify coupled rod-in-ring and rod-in-shell plasmonic systems with enhanced optical force and torque, including torque peaks shifted toward the near-infrared.
- Developing physics-inspired plasmonic optical-force descriptors to explain the shift mechanism and guide torque-enhanced applications such as molecular conformation stretching probes based on rod rotation.

Project: Thermally Switchable Multishell Plasmonic Nanomachines

- Designed concentric/eccentric Au-SiO2-DNA-Au multishell nanomachines whose two states exhibit distinct photothermal responses, hot-spot distributions, and bright and dark modes.
- Engineered DNA linkers with target melting temperatures so DNA dehybridization drives a concentric-to-eccentric transition, enabling on-off nanomachine control and downstream applications.

Professor Jianjun Cheng's Lab, Westlake University Hangzhou, China
Undergraduate Researcher **Sept. 2023 - Jan. 2025**

Project: Novel Rapid NCA Ring-Opening Polymerization Methods

- Synthesized BLG-NCA monomers and PBLG polypeptides, and quantified polymerization kinetics by in situ FTIR spectroscopy.
- Investigated macrodipole-assisted macroinitiators as a strategy to accelerate NCA ring-opening polymerization and improve polymerization efficiency.

International Genetically Engineered Machine (iGEM) Competition Hangzhou, China
Team Member **Dec. 2023 - Nov. 2024**

- Designed an E. coli genetic circuit for cancer cell killing and supported directed genetic modification of engineered bacteria.
- Engineered bacterial-surface cancer cell recognition receptors to improve the specificity of bacteria-mediated cancer therapy.
- Contributed to a project recognized with an International Silver Medal.

Awards and Honors

- National Scholarship (2025)
- Learning Innovation Award, Westlake University (2025)
- Outstanding Student, Westlake University (2025)
- Zhejiang Provincial Government Scholarship, Zhejiang Province (2024)
- Learning Innovation Award, Westlake University (2024)
- Outstanding Contribution Award, Student Ambassador Program, Westlake University (2024)
- Top Ten Singer Finalist, Westlake University (Jan. 2024)
- Best Singer Award and Best Performance Award, Undergraduate Music Concert (Jun. 2024)

Leadership & Activities

Head of the Academic Department, Student Development Center Hangzhou, China
Student Ambassador **Sept. 2024 - Jun. 2025**

- Organized faculty-student discussions on academic planning, research training, and campus life.
- Built and maintained an academic resource-sharing website for Westlake undergraduates.

Westlake University Student Ambassador Program Hangzhou, China
Student Ambassador **Sept. 2023 - Jun. 2024**

- Represented Westlake University at admissions outreach events, campus tours, and information sessions for prospective students.
- Coordinated student exchanges with ambassadors from the University of Nottingham Ningbo China.

Science Outreach Volunteer Hangzhou, China
Cancer Science Popularization for Primary and Middle Schools **Jul. 2024**

- Led interactive cancer biology and prevention sessions for primary and middle school students.

Campus Volunteer Activities Hangzhou, China
Westlake University Events Volunteer **Sept. 2023 - Present**

- Provided logistical support for major university events, including sports day, the Westlake Higher Education Forum, and Huxin Public Lectures.

Skills

Laboratory Technical Skills: polypeptide synthesis; basic organic synthesis; nanorod and nanoshell synthesis; TEM characterization

Computational Skills: deep learning model design (CNN, GNN, Transformer, DiT); high-throughput FDTD simulation; automated FDTD simulation-data extraction platform development

Language Skills: Native Chinese; Fluent English (IELTS 7.0)